

CONTENT-BASED IMAGE RETRIEVAL

Multimedia Databases (Exercises)

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Task 1: CBR

➤ What is CBR?

- A content-based retrieval system processes the information contained in data and creates an abstraction of its content in terms of attributes.
- **E.g.:** Online clothes shopping might allow users to search by traditional categories (brand, price range) and also in terms of visual attributes (color, texture)
- Steps:
 - Feature Extraction: The first step in the process is extracting image features to a distinguishable extent.
 - Matching: The second step involves matching these features to yield a result that is visually similar.



CBR

➤ Compared to each other, what are the benefits and limitations of ABIR and CBIR?

❑ Limitations of ABIR compared to CBIR

- Problem of Image annotations [Brahmi et al. 2004]
 - Large volumes of DB: manual image annotation is time-consuming and therefore costly.
 - The annotation is language dependent.
- Problem with human perception [Brahmi et al. 2004]
 - Human annotation is subjective
 - The accuracy and quality of the annotations are of the responsibility of the annotator and end-user.
- Problem in annotating images with words [Sclaroff et al. 1999]
 - Some images could not be annotated as it is difficult to describe their content with words.

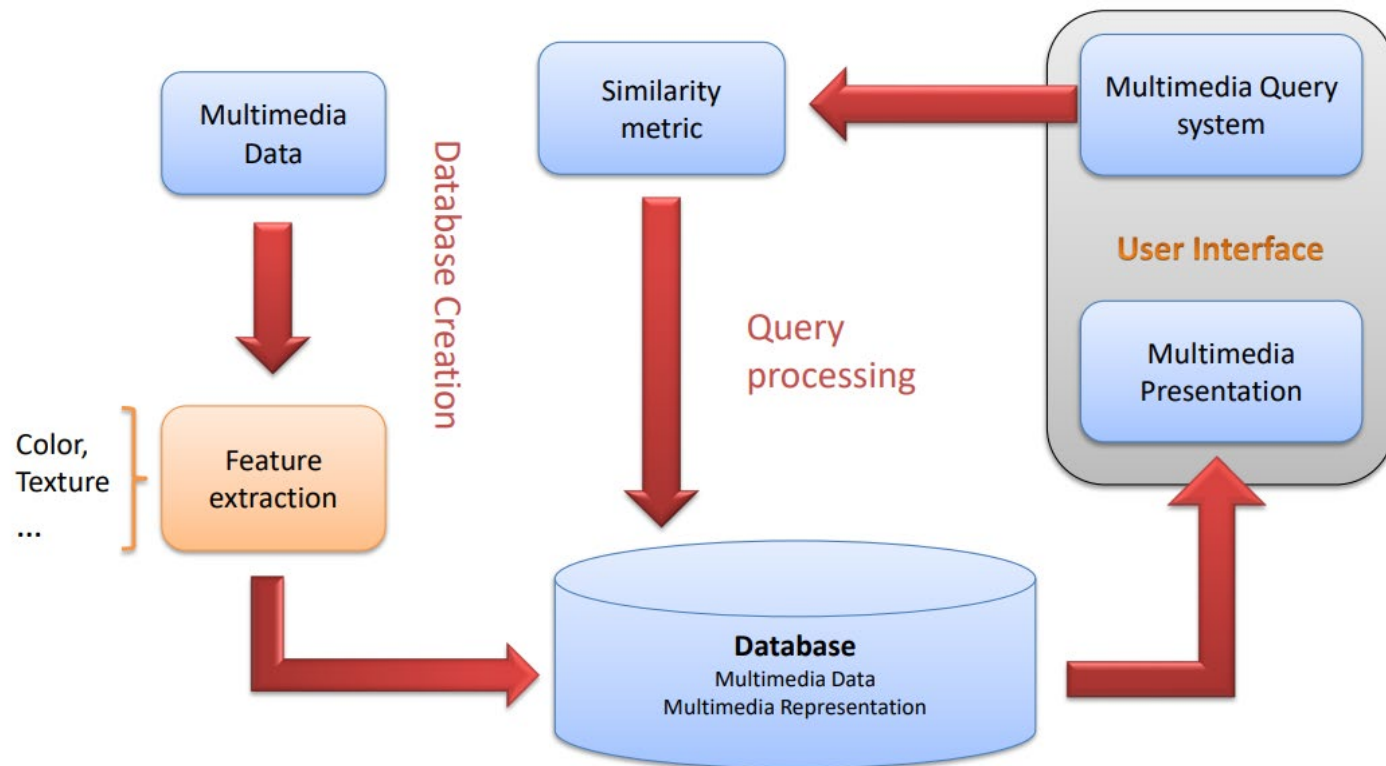
❑ Limitations of CBIR compared to ABIR

- Problem of Semantic Gap [Inoue et al. 2004, Brahmi et al. 2004]
 - Visual features cannot fully represent concepts.
- Problem of the availability of the image query [Inoue et al. 2004]
 - Users must have an example image in their hands

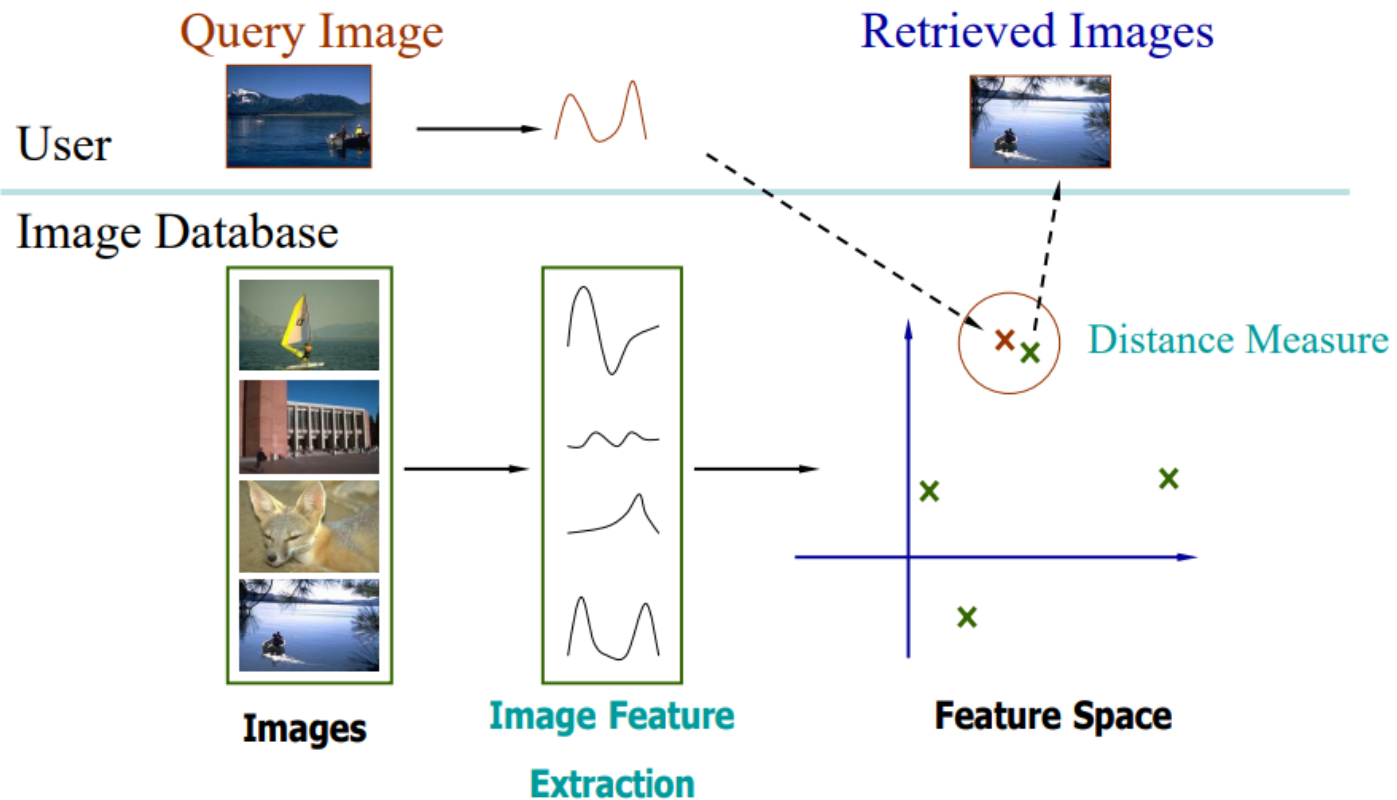


Task 1: CBR

- What are the components of a CBIR architecture?
Explain the basic principle of each component using an example?



Task 1: CBR



Task 1: CBR

➤ What is a feature vector?

- Numerical representation of object **features** in an n-dimensional **vector**
- E.g. feature = avg color, fv = (255,0,0)

➤ Which problem arise when indexing features vector?

- ‘The query performance of access methods decreases , if the dimensionality of the underlying data set becomes high.’
by Berchtold et al.
- A large number of features diminishes the distinguishing power of each feature and the indexing structures.

➤ ‘Curse of Dimensionality’

- Development of specific index structures that are specialized for high-dimensional data
- Reduction of the vector dimension:
 - Feature selection
 - Feature extraction



Task 2: CBR Definitions

1. Dominant Color

- The Dominant Color Descriptor allows specification of a small number of dominant color values + statistical properties like distribution and variance.
- In contrast to histogram color, only the representative colors are selected from each region or image.
- Its purpose is to provide an effective, compact and intuitive representation of colors present in a region or image.



Task 2: CBR Definitions

- **Dominant Color**

- The dominant color descriptor is defined by :

$$F = \{(c_i, p_i, v_i), s\}, (i = 1 \dots N)$$

- N is the total number of color clusters (= bins in the histogram) in the image region, such that $1 \leq N \leq 8$
- c_i is the dominant color vector (3d RGB vector),
- p_i is the percentage for each dominant color, such that
 - $P_i \in [0,1]$ and their sum is equal to 1,
- v_i is its color variance,
- s is a scalar that represents the overall spatial coherency of the dominant colors in the image.

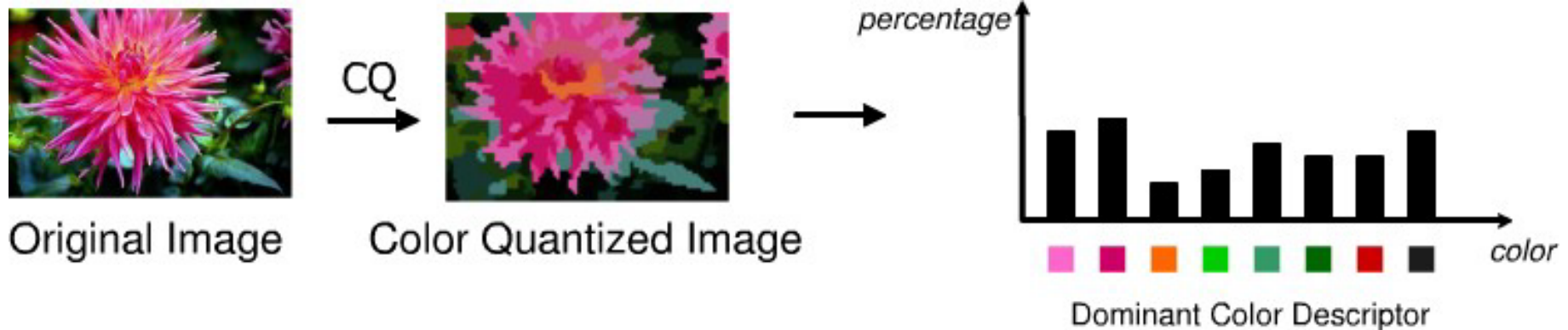


Task 2: CBR Definitions

• Dominant Color

- Often, the color variance v and spatial coherency s are not considered, which simplifies the definition of the dominant color descriptor to:

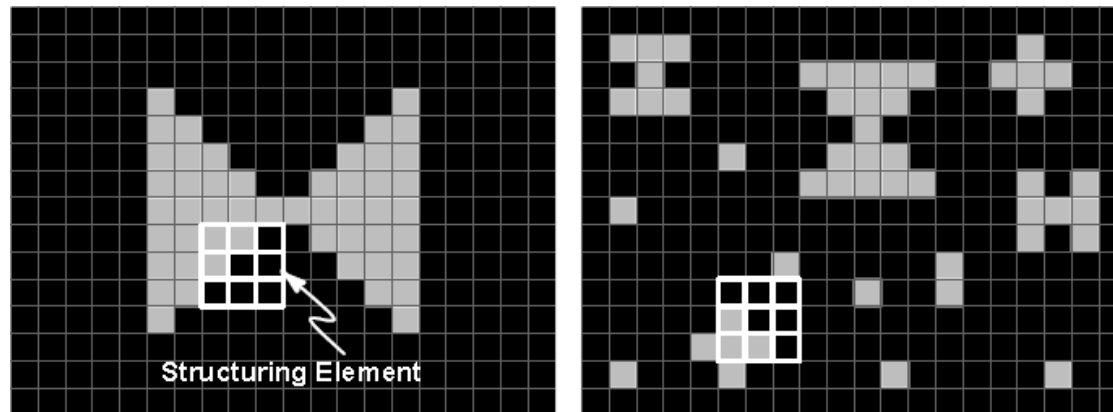
$$F = \{(c_i, p_i)\}, (i = 1 \dots N)$$



Task 2: CBR Definitions

2. Spatial Coherency

- The spatial coherency of a given dominant color is measured with **the normalized average number of connected pixels of this color**. It is computed using a 3x3 mask
- The overall spatial coherency is a linear combination of the individual spatial coherencies weighted with the corresponding percentages p_i .



Note: these two images have the same colors but the distribution of the colors is not the same.

Task 2: CBR Definitions

3. Distance metrics

- Degree of similarity between two points is measured by distance in data space. Several metrics have been established, i.e. euclidian distance.

4. Curse of Dimensionality

- CBR performs worse when dimensionality of the FVs increases.

5. Types of content-based queries

- **Point Query:** Retrieve all points with identical feature vector
- **Range Query:** Retrieve all points with a maximum distance from query point.
- **K-Nearest Neighbor Query:** Returns the k most similar results



Task 3: CBIR System

- What are the necessary conditions that must be fulfilled in order to be able to issue the following query to a CBIR system:
 - *Give me all images which contain a blue truck!*
- What are the problems that can occur?



Task 3: CBIR System

- **What are the necessary conditions that must be fulfilled in order to be able to issue the following query to a CBIR system:**
 - **Give me all images which contain a blue truck!**

To resolve this question using a "query by feature", we could:

1. Define the set of colors that correspond to the concept of "blue"
2. Define a set of contours (shape descriptors), which correspond to the shape of a truck.
3. Perform a query by features based on these two features.

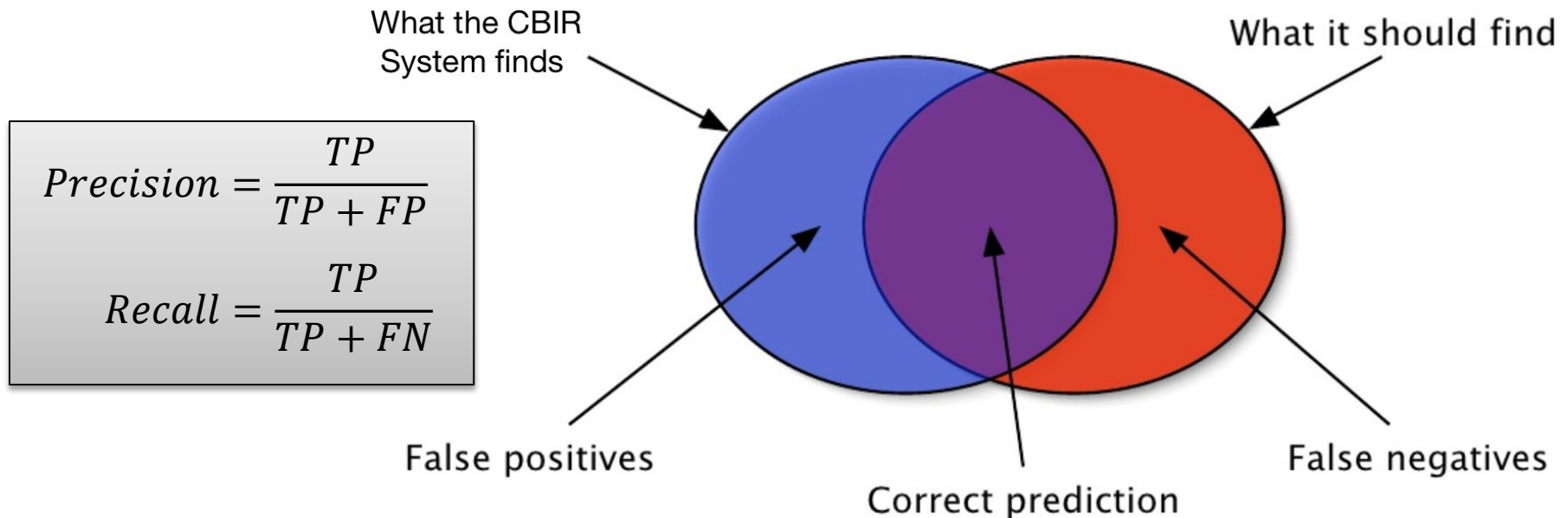
Task 3: CBIR System

- **What are the problems that can occur?**
- In the QBF-solution, the difficulty lies in the definition of the shape, as trucks can have very different contours (e.g. military trucks, commercial trucks, agricultural trucks etc.).
- **To which extent the definition of the shape should be precise?**
 - precise definition will lead to a good precision but a bad recall.
 - less precise definition will provide bad precision (false positives) but a better recall.
- Thus, the appropriate strategy depends on the exact goals of the query.
 - **But, it is very difficult to get this information from the user.**



Task 3: CBIR System

- How good is the result?



Precision How many of the returned entities are relevant?

High precision = few false positives

Recall How many relevant entities are returned?

High recall = few false negatives

Task 4: Image Indexing by Colours

- Apply a uniform colour quantization of 8 bins.
- Which quantization area (range) do the colours in the two images belong to?

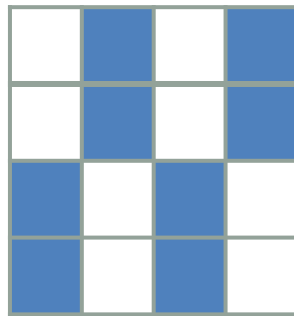


Image 1

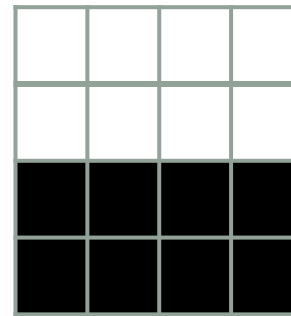
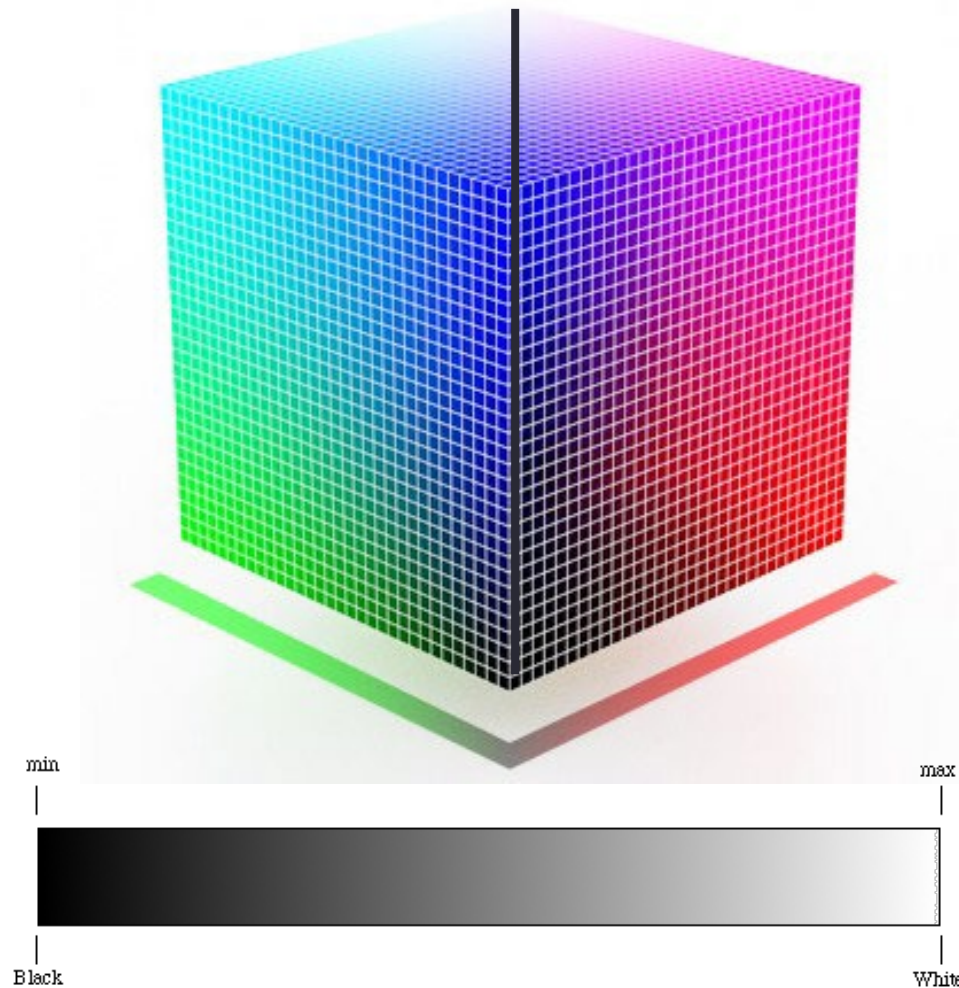
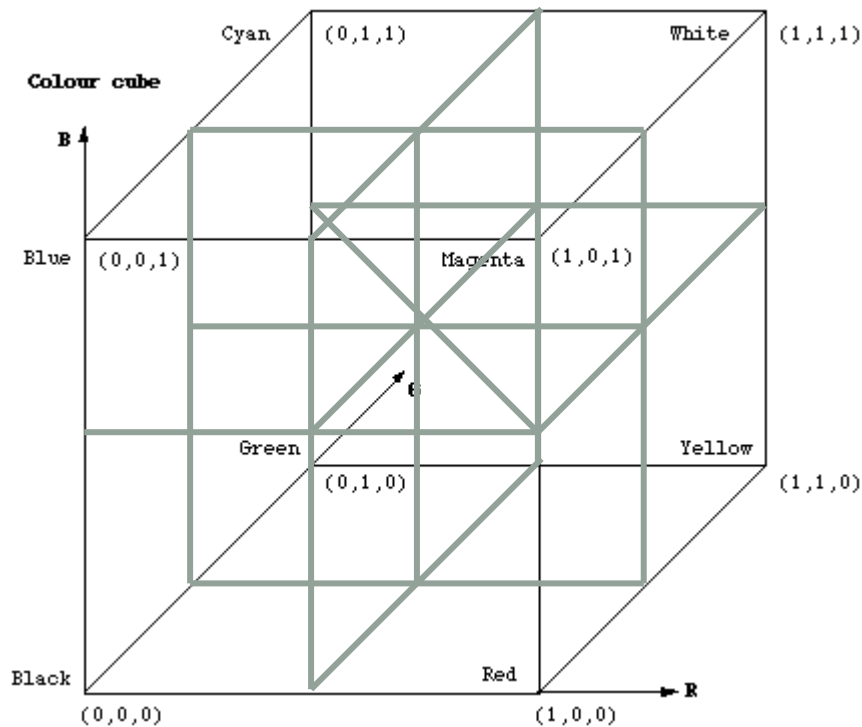


Image 2

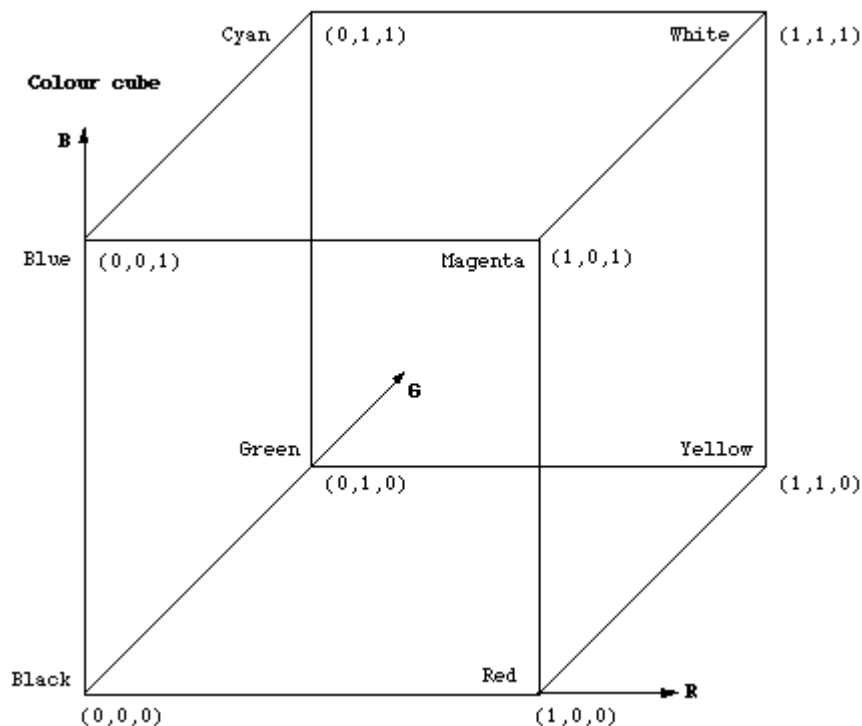
Task 4: Image Indexing by Colours

- Apply a uniform colour quantization of 8 bins.



Task 4: Image Indexing by Colours

- Apply a uniform colour quantization of 8 bins.



- | | |
|-----------|-----------|
| ▪ (0,0,0) | ▪ (1,0,0) |
| ▪ (0,0,1) | ▪ (1,0,1) |
| ▪ (0,1,0) | ▪ (1,1,0) |
| ▪ (0,1,1) | ▪ (1,1,1) |

Task 4: Image Indexing by Colours

- Which quantization area (range) do the colours in the two images belong to?

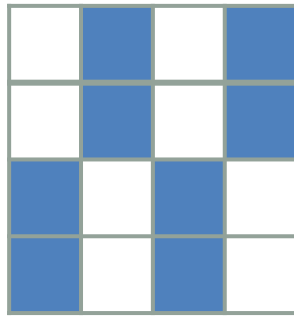


Image 1

- Blue pixels are part of bin (0,0,1)
- White pixels are part of bin (1,1,1)

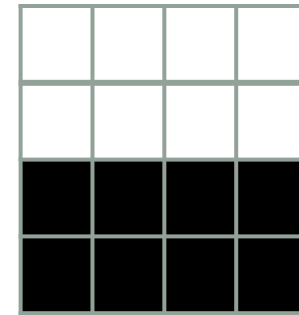


Image 2

- Black pixels are part of bin (0,0,0)
- White pixels are part of bin (1,1,1)

Task 4: Image Indexing by Colours

- Create a color histogram for both images

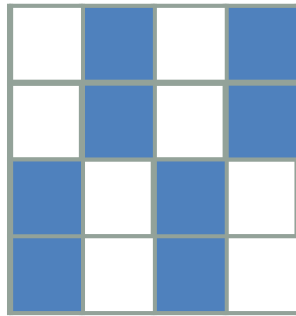


Image 1

- $H_1 = (0,0,0,0,8,0,0,8)$

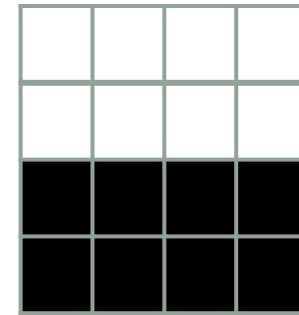


Image 2

- $H_2 = (8,0,0,0,0,0,0,8)$

Task 4: Image Indexing by Colours

- Apply a uniform bin quantization for 2 bits (max value quantization and mask quantization)

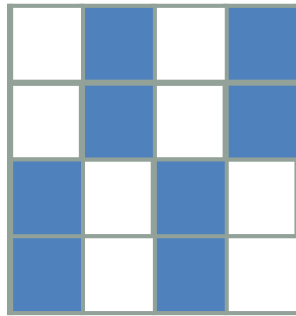


Image 1

$$\blacksquare H_1 = (0,0,0,0,8,0,0,8)$$

$$\Rightarrow H_1 = (0,0,0,0,3,0,0,3)$$

Or by using a 1100 AND mask on 1000

$$\Rightarrow H_1 = (0,0,0,0,2,0,0,2)$$

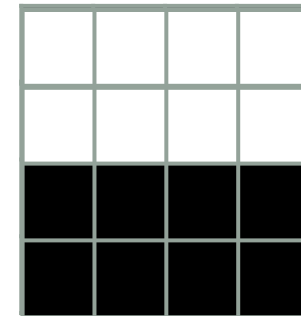


Image 2

$$\blacksquare H_2 = (8,0,0,0,0,0,0,8)$$

$$\Rightarrow H_2 = (3,0,0,0,0,0,0,3)$$

Or by using a 1100 AND mask on 1000

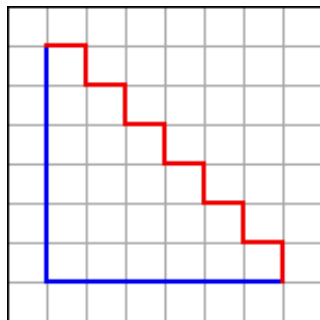
$$\Rightarrow H_1 = (0,0,0,0,2,0,0,2)$$

Task 5: Similarity of images because of colour distribution

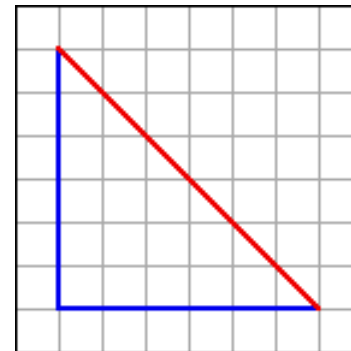
- **Minkowski Distances**

Use the two images of exercise 4 and the results of exercise 4.2 (without bin quantification). Determine the similarity of the images with the help of Minkowski distances:

- L_1 : Manhattan distance
- L_2 : Euclidian distance
- L_∞ : Maximal distance (also called Tschebyschow distance)



Manhattan Distance



Euclidean Distance

Task 5: Similarity of images because of colour distribution

- **Minkowski Distances**

Starting with $P = (x_1, x_2, \dots, x_n)$ and $Q = (y_1, y_2, \dots, y_n) \in R^n$

$$L_p(P, Q) = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{1/p}$$

▪ $H_1 = (0, 0, 0, 0, 8, 0, 0, 8)$

▪ $H_2 = (8, 0, 0, 0, 0, 0, 0, 8)$

$$L_1(H_1, H_2) = |0-8| + |0-0| + |0-0| + |0-0| + |8-0| + |0-0| + |0-0| + |8-8| = 8+8 = 16$$

$$L_2(H_1, H_2) = \sqrt{(0-8)^2 + \dots + (8-0)^2 + \dots + (8-8)^2} = \sqrt{128} \approx 11,3$$

$$L_\infty(H_1, H_2) = \max_i |x_i - y_i| = 8$$

Task 5: Similarity of images because of colour distribution

- **Statistical Distances**

$$H_1 = (5,5,5,5) \text{ and } H_2 = (8,5,4,3)$$

$$KS(P, Q) = \max_i |F^r(i, P) - F^r(i; Q)|$$

$F^r(i, P)$ is equivalent to the cumulative histogram of P in place i

Cumulative $H_1 = (5, 10, 15, 20)$

Cumulative $H_2 = (8, 13, 17, 20)$

$$\Rightarrow KS(H_1, H_2) = 3$$



Task 5: Similarity of images because of colour distribution

- **Chi-squared Distance**

$$H_1 = (5,5,5,5) \text{ and } H_2 = (8,5,4,3)$$

$$D_{\chi}(P, Q) = \sum_i \frac{(x_i - f'(i))^2}{f'(i)} \quad \text{with} \quad f'(i) = \frac{x_i + y_i}{2}$$

$$f'(1) = \frac{5 + 8}{2} = \frac{13}{2} \quad f'(2) = 5 \quad f'(3) = \frac{9}{2} \quad f'(4) = 4$$

.....

$$D_{\chi}(H_1, H_2) \approx 1.303$$

